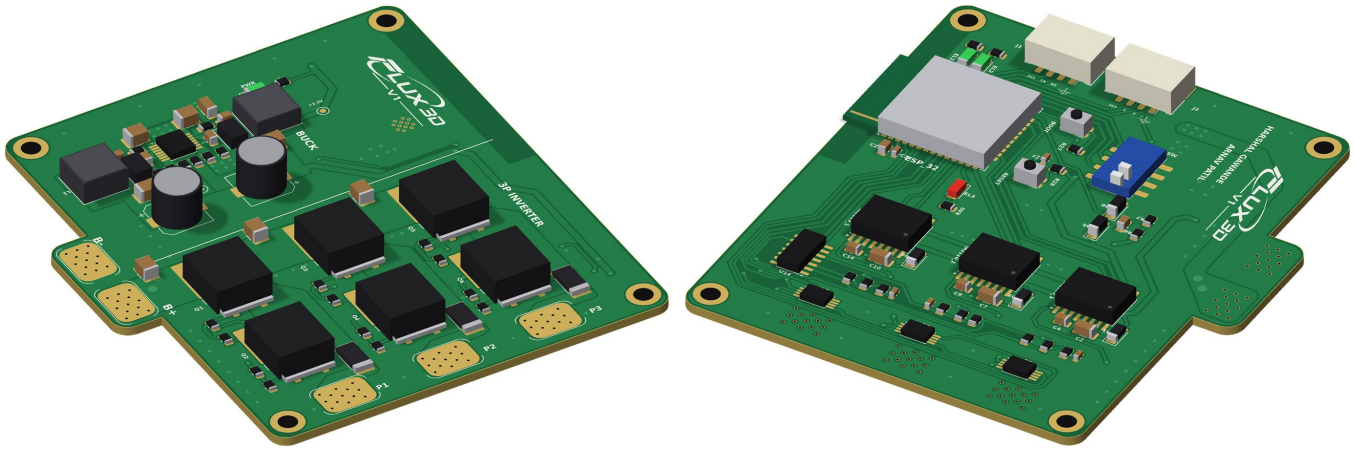
BLDC Controller v1

ESP32-based three-phase brushless DC motor controller board

14 – 25.2 V (4S – 6S) · 6 × 75 V MOSFET inverter · ±80 A current sensing · sensorless BEMF · Wi-Fi / Bluetooth

79.4 × 88.3 mm · 2-layer 1.6 mm PCB · 4 × M3 mounting

Designed by Harshal Gawande & Arnav Patil
Designed and owned by Harshal Gawande and Arnav Patil



Top (power) side and bottom (control) side — illustrative 3D renders generated from the PCB geometry

1 Features

- **Three-phase inverter:** 6 × IRFS3607 N-MOSFETs (75 V, 80 A, 7.34 mΩ typ.) in D2PAK, 3 × IR2110S half-bridge gate drivers with bootstrap
- **Inline phase-current sensing:** 3 × 1 mΩ shunts + 3 × INA240A1 (20 V/V, bidirectional) → ±80 A measurement range
- **Sensorless operation:** back-EMF comparator network (LMV324) on all three phases; DC-bus voltage monitor
- **Integrated supplies:** TPS54383 dual buck derives the 12 V gate-drive rail and the 3.3 V logic rail directly from VBUS
- **ESP32-WROOM-32E** (4 MB flash, Wi-Fi + Bluetooth, on-board antenna); UART header (J1), I/O header (J2), BOOT/RESET buttons, DIP switch, 4 LEDs
- **Large wire pads** (6.6 × 9 mm) for battery and motor leads; 4 × M3 mounting holes; 2 test pads (+12 V, +3.3 V)

Applications

- Robotics and AGV drive wheels
- Electric skateboards, scooters and light e-mobility (≤ 6S)
- Brushless pump, fan and gimbal drives
- BLDC / PMSM firmware development (six-step or FOC)

Device summary

Key parameter	Value
Supply (VBUS)	14 – 25.2 V recommended (4S – 6S Li-ion/LiPo); 30 V abs. max
Phase-current sense range	±80 A (1 mΩ shunt, INA240A1 20 V/V)
Power stage	6 × IRFS3607 (75 V / 80 A / 7.34 mΩ typ.)
Gate drive	3 × IR2110S, 12 V, 22 Ω gate resistors
Controller	ESP32-WROOM-32E-N4 (240 MHz, 4 MB flash)
Logic rail / gate rail	3.3 V / 12 V from on-board TPS54383
Board size	79.38 × 88.26 mm (incl. tab), 1.6 mm
Overall height	≈ 12.5 mm (top 6.5, PCB 1.6, bottom 4.4)
Layers / mounting	2 layers · 4 × M3 (Ø3.2 mm)
Interfaces	UART (J1), I/O (J2), BOOT, RESET, 3-bit DIP
Temperature (module-limited)	−40 °C to +85 °C ambient

Description

BLDC Controller v1 is a compact, two-layer motor-control board that combines a six-MOSFET inverter, gate drivers, per-phase current sensing, back-EMF sensing and an ESP32 module on one PCB. The power stage and buck converter sit on the top side; the ESP32, gate drivers, sense amplifiers and connectors sit on the bottom side. Motor commutation is implemented entirely in firmware running on the ESP32.

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2 Specifications

How to read these numbers. This datasheet was generated from the KiCad schematic/PCB and the ordering BOM. Ratings are the *limits of the weakest fitted component* or values calculated from the schematic; they are not measurements of a built board. Always confirm against the component datasheets and validate current/thermal limits on your own hardware before relying on them.

2.1 Absolute maximum ratings

Parameter	Min	Max	Unit	Limited by
VBUS (B+ to B−)	0	30	V	TPS54383 VIN abs. max (28 V recommended max.). No reverse-polarity or surge protection on board.
Voltage on J1 / J2 pins and ESP32 I/O	−0.3	3.6	V	ESP32-WROOM-32E supply abs. max
Phase current — measurable	−80	+80	A	INA240A1 range with 1 mΩ shunt (ideal ±82 A)
Phase current — shunt power limit (continuous)	—	≈ 55	A	2512 shunt, 3 W rating: √(3 W / 1 mΩ); no derating applied
Operating ambient temperature	−40	+85	°C	ESP32-WROOM-32E-N4 module rating

Stresses beyond these may cause permanent damage. The board contains ESD-sensitive devices; handle with standard ESD precautions.

2.2 Recommended operating conditions

Parameter	Min	Typ	Max	Unit	Notes
VBUS supply voltage	14	22.2	25.2	V	4S (16.8 V full) to 6S (25.2 V full). The minimum keeps the 12 V gate-drive rail in regulation (approx.).
12 V gate-drive rail (TP1)	—	12.0	—	V	Generated on board; IR2110 VCC range is 10–20 V
3.3 V logic rail (TP2)	—	3.3	—	V	Generated on board; powers ESP32, INA240, LMV324, IR2110 logic
PWM frequency (firmware-defined)	—	10 – 50	500	kHz	500 kHz is the IR2110 limit; practical limit set by MOSFET switching loss and layout
Logic-level PWM inputs	0	—	3.3	V	Driven directly by ESP32 GPIO (VDD of IR2110 = 3.3 V)
Ambient temperature	−40	25	85	°C	Derate current with temperature; see §2.4

2.3 Electrical characteristics (calculated, T_A = 25 °C)

Parameter	Conditions / basis	Value	Unit
12 V rail setpoint	V_{FB} 0.8 V · R10 51.1 kΩ / R11 3.65 kΩ	12.0	V
3.3 V rail setpoint	V_{FB} 0.8 V · R8 20 kΩ / R9 6.34 kΩ (BOM; schematic shows 6.4 kΩ → 3.30 V)	3.32	V
Buck output rating	TPS54383, per channel (practical limit set by L1 ≥ 1 A, L2 2.5 A per BOM)	3	A
Gate resistors	R14, R15, R19, R20, R23, R24; gate–source pull-downs 10 kΩ	22	Ω
Bootstrap capacitors	C2, C7, C10 (VB–VS)	1	µF
Current-sense gain	INA240A1	20	V/V
Current-sense sensitivity	20 V/V × 1 mΩ	20	mV/A
Zero-current output	REF1 = GND, REF2 = 3.3 V (mid-supply reference)	1.65	V
Current resolution	3.3 V / 4096 LSB ÷ 20 mV/A (12-bit ESP32 ADC, before ADC non-linearity)	≈ 40	mA/LSB
Shunt dissipation	I^2 × 1 mΩ at 20 A / 40 A / 55 A	0.4 / 1.6 / 3.0	W

Parameter	Conditions / basis	Value	Unit
MOSFET conduction loss	Two FETs conduct per step; $R_{DS(on)}$ 7.34 mΩ typ., 25 °C, at 20 A (hot $R_{DS(on)}$ is higher)	≈ 5.9	W
VBUS monitor ratio	R6 56 kΩ / R7 4.7 kΩ (clamped by D5/D6)	0.0774 (1.95 V @ 25.2 V)	—
BEMF divider ratio (as built)	R33/35/37 33 kΩ over R32/34/36 10 kΩ → 5.9 V at 25.2 V (see §11, item 2)	0.2326	—
PWR LED current	(3.3 V − 2.0 V) / 330 Ω	≈ 4	mA

2.4 Current handling and thermal notes

The board has not been thermally characterised. Continuous current is limited by PCB copper, MOSFET cooling and the shunt rather than by the MOSFET ID rating. The power paths use copper pours on both layers, 1.0 mm top-side tracks and 3.0 mm bottom-side tracks; copper weight is not specified in the design files (see §9). For sustained high current use 2 oz (70 µm) outer copper, airflow or a heat-sink on the D2PAK tabs, and verify the temperature rise at your operating point.

3 Functional block diagram

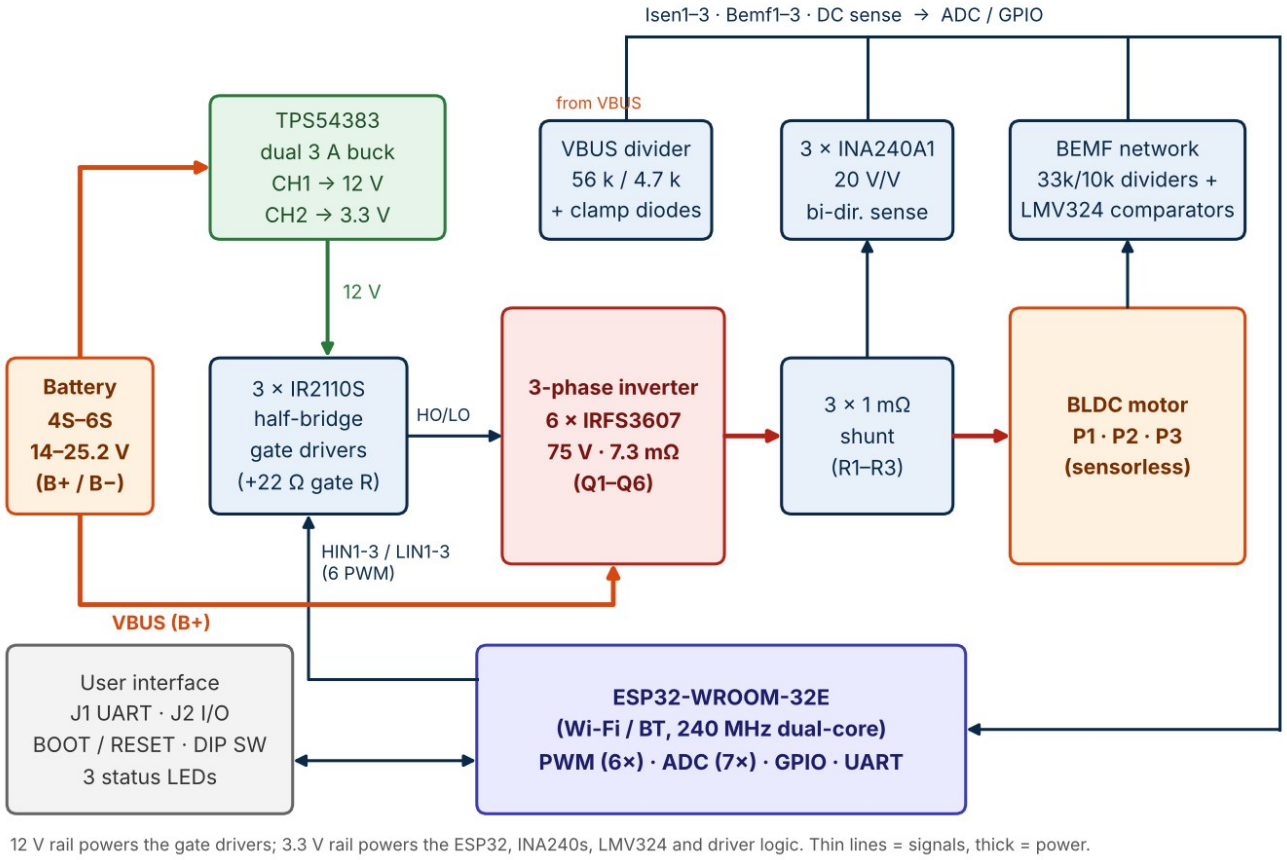


Figure 1 — Functional block diagram

VBUS feeds the inverter directly and the two buck channels of the TPS54383. The 12 V channel supplies the three IR2110S gate drivers; the 3.3 V channel supplies the ESP32, current-sense amplifiers, comparators and driver logic. The ESP32 generates six PWM signals (HIN/LIN per phase) and reads back three phase currents, three back-EMF comparator outputs and the DC-bus voltage.

4 Pin configuration and functions

The board exposes five high-current wire pads, two 4-pin JST-PH headers, two push-buttons, a DIP switch, LEDs and test pads. Pad positions are given from the bottom-left corner of the board outline in the top view (X → right, Y → up). Net names are taken from the PCB netlist.

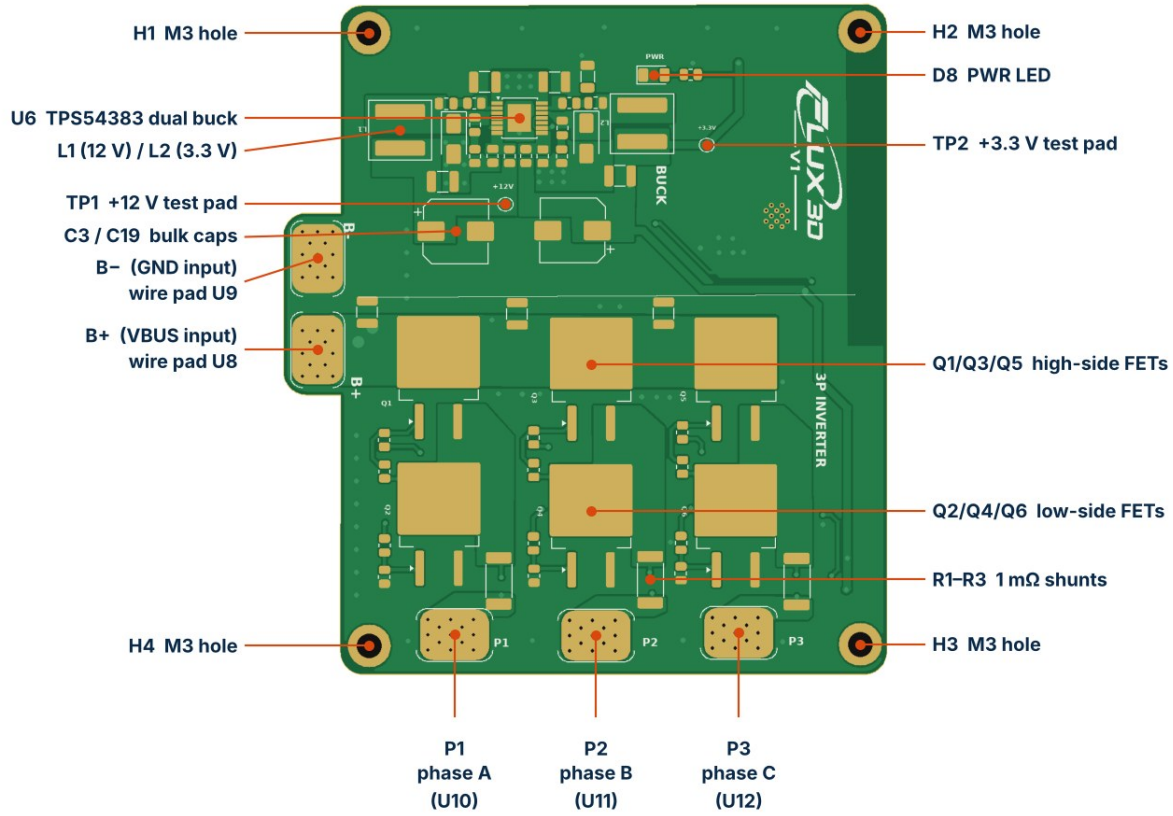


Figure 2a — Top side interface and key-component map

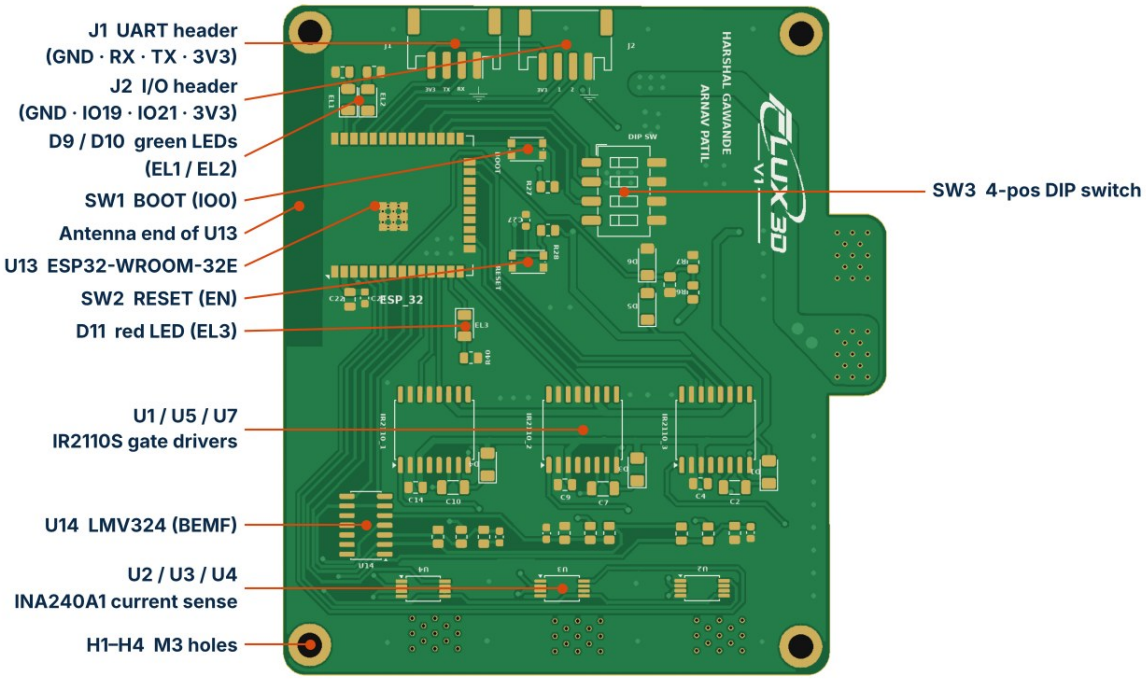


Figure 2b — Bottom side interface and key-component map (mirrored, as seen when the board is flipped)

4.1 Power terminals (wire pads, top side)

Label	Ref	Net	Function	Pad (X × Y)	Centre (mm)
B+	U8	VBUS	Battery positive input (14 – 25.2 V)	6.6 × 9.0 mm	4.50, 42.75
B−	U9	GND	Battery negative / power ground	6.6 × 9.0 mm	4.50, 54.75
P1	U10	PR1	Motor phase A output (through shunt R1)	9.0 × 6.6 mm	22.50, 5.25
P2	U11	PR2	Motor phase B output (through shunt R2)	9.0 × 6.6 mm	41.14, 5.15
P3	U12	PR3	Motor phase C output (through shunt R3)	9.0 × 6.6 mm	59.92, 5.51

Each pad is a 6.6 × 9 mm exposed-copper area with 13 plated stitching holes (Ø0.4 mm) tied to the matching copper pour on the bottom layer. Solder cables or lugs directly; pre-tin generously. The pads are exposed on the top side only.

4.2 J1 — UART header (bottom side, JST-PH 4-pin, 2.0 mm, SMD side-entry)

Pin	Silk	Net	ESP32 pin	Direction (board side)	Notes
1	GND	GND	—	—	Ground
2	RX	RXD0	GPIO3 (U0RXD)	Input to board	Connect to adapter TX
3	TX	TXD0	GPIO1 (U0TXD)	Output from board	Connect to adapter RX; also firmware-flash port
4	3V3	+3.3V	—	Power out	Board 3.3 V rail — do not back-power from adapter

4.3 J2 — I/O header (bottom side, JST-PH 4-pin, 2.0 mm, SMD side-entry)

Pin	Silk	Net	ESP32 pin	Notes
1	GND	GND	—	Ground
2	2	IO19	GPIO19	General-purpose I/O (BOM: I²C/aux)
3	1	IO21	GPIO21	General-purpose I/O (BOM: I²C/aux); default I²C SDA pin on ESP32
4	3V3	+3.3V	—	Board 3.3 V rail

Function of J2 is defined by firmware. The pin order printed on the PCB (3V3 – 1 – 2 – GND) is the reverse of the pin numbering above.

4.4 Switches, LEDs and test points

Ref	Name	Connection	Description
SW1	BOOT	GPIO0 ↔ GND (R27 10 kΩ pull-up to 3.3 V)	Hold during reset to enter the serial bootloader
SW2	RESET	EN ↔ GND (R28 10 kΩ pull-up, C27 1 µF)	Resets the ESP32
SW3	DIP SW	Positions 1/2/3 → GPIO17 / GPIO16 / GPIO4, other side GND; position 4 unconnected	Closed = pin pulled to GND (enable internal pull-ups in firmware)
D8	PWR	3.3 V → R18 330 Ω → D8 → GND	Green power indicator (always on when 3.3 V present)
D9, D10	LED3, LED2	GPIO22, GPIO23 (see §11, item 1)	Green user LEDs (silk EL-series legends)
D11 (PCB “EL3”)	LED1	GPIO14 (see §11, item 1)	Red user LED
TP1	+12 V	12 V rail	Test pad Ø0.8 mm, top side — position 29.22, 61.87
TP2	+3.3 V	3.3 V rail	Test pad Ø1.0 mm, top side — position 55.70, 69.67

4.5 ESP32-WROOM-32E pin assignment (U13)

Module pin	GPIO	Net / signal	Connected to	Notes
2	3V3	+3.3V	3.3 V rail	C20, C22, C24 decoupling nearby
1, 15, 38, 39	GND	GND	Ground	
3	EN	EN	SW2, R28, C27	Reset / chip enable
5	GPIO39 (VN)	Isen1	U2 INA240 OUT (phase 1)	ADC1, input-only
4	GPIO36 (VP)	Isen2	U3 INA240 OUT (phase 2)	ADC1, input-only
6	GPIO34	Isen3	U4 INA240 OUT (phase 3)	ADC1, input-only
7	GPIO35	DC sense	VBUS divider R6/R7 (+D5, D6, C11)	ADC1, input-only
8	GPIO32	Bemf1	U14A comparator — phase P3	ADC1 / digital
10	GPIO25	Bemf2	U14B comparator — phase P2	ADC2: not usable for analog reads while Wi-Fi is active
9	GPIO33	Bemf3	U14C comparator — phase P1	ADC1 / digital
24	GPIO2	HIN1	U1 HIN (phase 1 high-side)	Strapping pin
23	GPIO15	LIN1	U1 LIN (phase 1 low-side)	Strapping pin
16	GPIO13	HIN2	U5 HIN (phase 2 high-side)	
14	GPIO12	LIN2	U5 LIN (phase 2 low-side)	Strapping pin (flash voltage)
12	GPIO27	HIN3	U7 HIN (phase 3 high-side)	
11	GPIO26	LIN3	U7 LIN (phase 3 low-side)	
25	GPIO0	BOOT	SW1, R27	Strapping pin (boot mode)
28 / 27 / 26	GPIO17 / 16 / 4	DIP1 / DIP2 / DIP3	SW3 positions 1 / 2 / 3	
13 / 37 / 36	GPIO14 / 23 / 22	LED1 / LED2 / LED3	D11 / D10 / D9 cathodes	See §11, item 1
34 / 35	GPIO3 / GPIO1	RXD0 / TXD0	J1 pins 2 / 3	UART0
31 / 33	GPIO19 / GPIO21	IO19 / IO21	J2 pins 2 / 3	
29, 30	GPIO5, GPIO18	—	Not connected	Free, not routed
17–22, 32	NC	—	Not connected	Module NC pins

4.6 Per-phase signal chain

Phase	Motor pad	Driver	High / low FET	Shunt	Current amp → ADC	BEMF comparator → GPIO	HIN / LIN GPIO	Bootstrap
1	P1 (U10)	U1	Q1 / Q2	R1	U2 → GPIO39	U14C → GPIO33	GPIO2 / GPIO15	D1, C2
2	P2 (U11)	U5	Q3 / Q4	R2	U3 → GPIO36	U14B → GPIO25	GPIO13 / GPIO12	D3, C7
3	P3 (U12)	U7	Q5 / Q6	R3	U4 → GPIO34	U14A → GPIO32	GPIO27 / GPIO26	D4, C10

IR2110S pin use (all three drivers): VCC = 12 V, VDD = 3.3 V, SD tied to GND (hardware shutdown not used), VSS/COM = GND, VS = phase node, VB = bootstrap node.

5 Detailed description

5.1 Power input and DC link

VBUS enters at B+ / B−. On-board DC-link capacitance is 2 × 10 µF / 50 V X7S (C12, C13) plus 3 × 2.2 µF / 100 V X7R (C23, C25, C26) placed at the half-bridges. There is **no bulk electrolytic capacitor on VBUS**; add low-ESR bulk capacitance (for example 470 µF, ≥ 35 V) close to the B+/B− pads, together with a fuse or pre-charge/inrush limiter and a battery switch.

5.2 Power supplies (U6, TPS54383)

Both channels of the TPS54383 run from VBUS. Channel 1 (L1 47 µH, D2 SS34) generates 12 V for the gate drivers and the bootstrap supply; channel 2 (L2 22 µH, D7 SS34) generates 3.3 V. Each switch node has an RC snubber (10 Ω + 1 nF) and a boot capacitor (33 nF with 3.3 Ω). The enable pins are tied to GND in the schematic, so both rails start as soon as VBUS is present. Because the 12 V rail is a step-down from VBUS, the board needs VBUS ≳ 14 V for a regulated gate-drive supply.

5.3 Gate drivers and inverter (U1, U5, U7, Q1–Q6)

Each phase leg is driven by one IR2110S with a bootstrap diode (D1/D3/D4) and 1 µF bootstrap capacitor (C2/C7/C10). Logic inputs HIN/LIN are driven directly from ESP32 GPIOs at 3.3 V. Gate resistors are 22 Ω with 10 kΩ gate-source pull-downs. The IR2110 has **no dead-time generator or shoot-through interlock**: the firmware must insert dead-time between complementary switching edges and must keep both inputs of each driver low during reset and start-up.

5.4 Current sensing (R1–R3, U2–U4)

A 1 mΩ 2512 shunt is placed in series with each motor phase between the inverter node (P*n*) and the motor pad (PR*n*). An INA240A1 measures it differentially and is configured for bidirectional output: REF1 = GND and REF2 = 3.3 V place zero current at about 1.65 V. Sensitivity is 20 mV/A, giving ±82 A ideal (±80 A quoted) full scale. Phase-current samples are valid when the inverter low-side or high-side path through that phase is conducting; INA240 enhanced PWM rejection allows sampling during switching.

5.5 Back-EMF sensing (U14, LMV324)

Each phase node is divided by 33 kΩ / 10 kΩ (R33/R32, R35/R34, R37/R36) with a 1 nF filter capacitor (C28–C30) and compared against a virtual neutral formed by R29–R31 (33 kΩ). The three comparator outputs (Bemf1–3) go to ESP32 GPIO32, GPIO25 and GPIO33 for zero-crossing detection in sensorless six-step control. See §11 item 2 regarding the divider ratio.

5.6 Controller and boot behaviour (U13)

The ESP32-WROOM-32E runs from the 3.3 V rail. The module antenna end overhangs the board edge by ≈ 0.7 mm and the area around it is kept free of copper on the top layer; keep metal, cables and batteries away from it. Strapping pins GPIO0, GPIO2, GPIO12 and GPIO15 are shared with BOOT and the HIN1/LIN1/LIN2 gate-driver inputs, so gate-driver input levels at reset can influence boot mode (GPIO12 selects flash voltage and must be low at reset). Hold SW1 (BOOT) low and tap SW2 (RESET) to enter the serial bootloader.

6 Application information

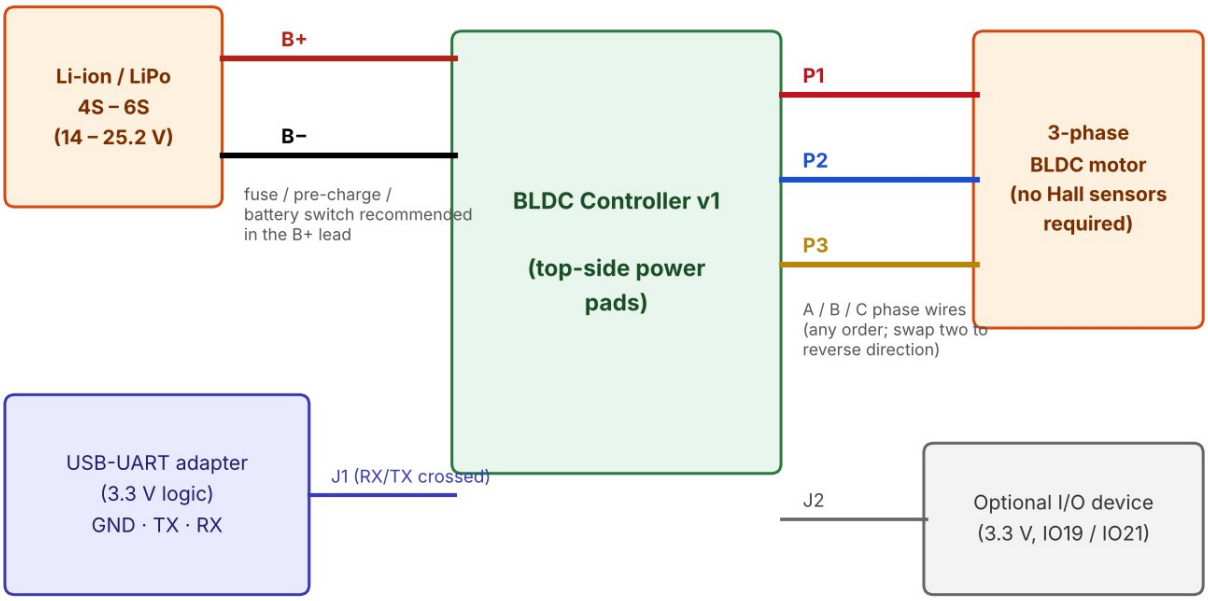


Figure 3 — Typical connection

6.1 Bring-up checklist

- Fit/verify all parts, apply the rework items in §11, and add external bulk capacitance across B+/B−.
- Connect the motor **without** a load on the shaft; wire a current-limited bench supply (start ≈ 14 – 16 V, a few amps limit) to B+/B−.
- Check TP1 = 12 V and TP2 = 3.3 V; the PWR LED (D8) should light.
- Connect a 3.3 V USB-UART adapter to J1 (GND–RX–TX crossed). Hold BOOT, tap RESET, flash firmware, then reset.
- Verify with a scope that HIN/LIN stay low at reset, then validate dead-time and phase-current readings before increasing the supply.

6.2 Recommended external parts and protection

- **Bulk capacitance** across B+/B−: low-ESR electrolytic, e.g. 470 µF ≥ 35 V (the earlier BOM used 2 × 470 µF / 35 V); mount close to the wire pads.
- **Battery protection:** fuse or circuit breaker, anti-spark/pre-charge connector and reverse-polarity protection — none of these is on the board.
- **Surge clamp:** a 30 V-class TVS (BOM review suggests SMBJ26A) or an input filter near U6, because VBUS is limited to 30 V by the buck regulator.
- **Wiring:** size battery and phase cables for the intended current; pre-tin the pads and strain-relieve the leads away from the ESP32 antenna end.
- **Cooling:** airflow or a heat-sink on the D2PAK tabs for sustained currents (§2.4).

6.3 Firmware notes

- Generate the six PWM signals on the HIN/LIN GPIOs in §4.5 (any ESP32 GPIO can be routed to a PWM peripheral) with complementary outputs and software dead-time.
- Calibrate each current channel at zero current (≈ 1.65 V) at start-up; sample with ADC1 (Isen1–3, DC sense). Bemf2 (GPIO25) is on ADC2 — treat it as a digital input when Wi-Fi is used.
- DIP switch inputs close to GND: enable the internal pull-up on GPIO17, GPIO16 and GPIO4.
- Keep GPIO0/2/12/15 in a safe state during boot; see §5.6.

7 Mechanical data

TOP VIEW (component side F.Cu)

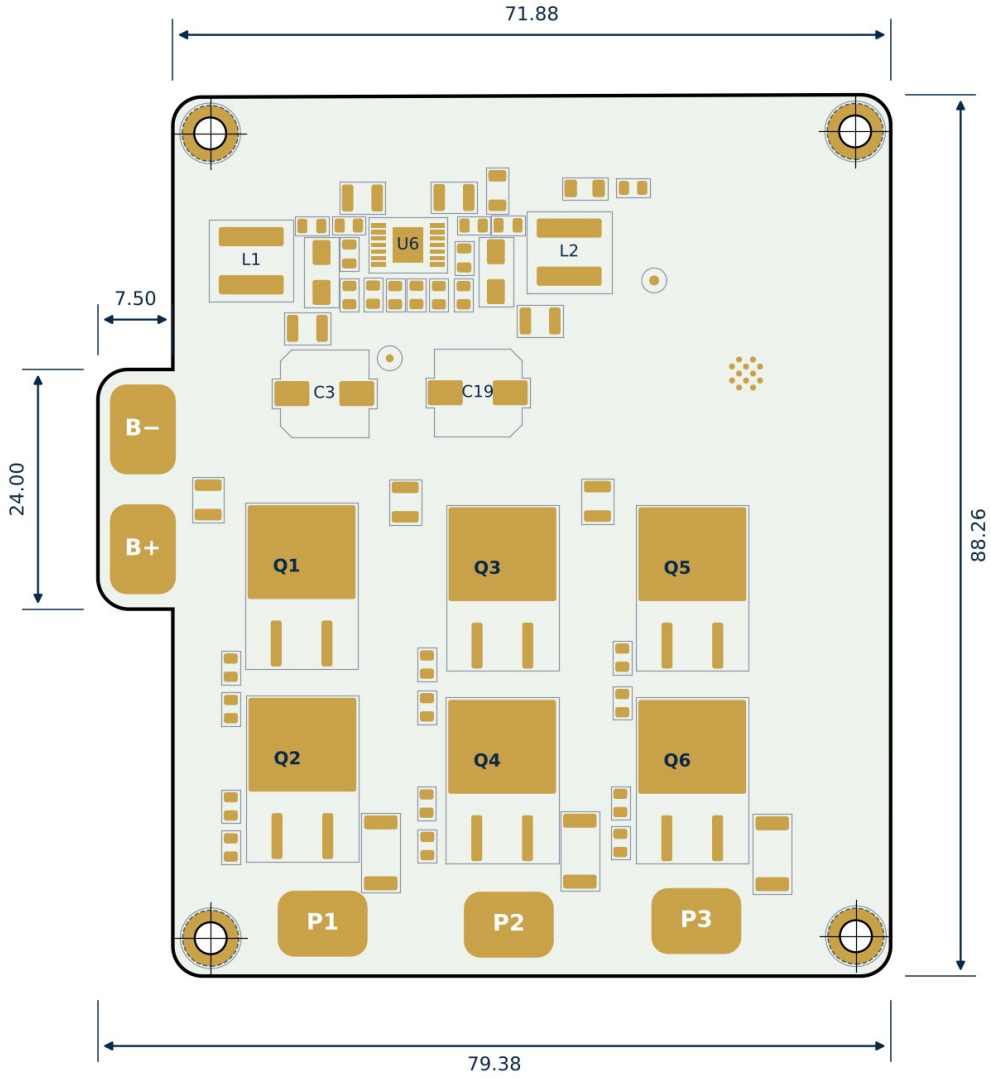


Figure 4a — Top view dimensions (mm); origin at the bottom-left corner of the board bounding box

7.1 Board dimensions

Feature	Value	Notes
Overall outline (bounding box)	79.38 × 88.26 mm	Area of outline ≈ 65.0 cm²
Main body	71.88 × 88.26 mm	Corner radii ≈ 3 mm
Left tab (B+ / B− pads)	7.50 × 24.00 mm	Offset 36.75 mm from bottom edge, 27.51 mm from top edge
PCB thickness	1.6 mm	Two copper layers
Component height — top side	≈ 6.5 mm above PCB	Tallest: C3, C19 (Ø8 × 6.5 mm cans per BOM); footprint allows up to 10 mm
Component height — bottom side	≈ 4.4 mm below PCB	Tallest: J1, J2 (estimated); ESP32 module 3.2 mm
Overall thickness	≈ 12.5 mm	With 10 mm tall capacitors: ≈ 16.0 mm

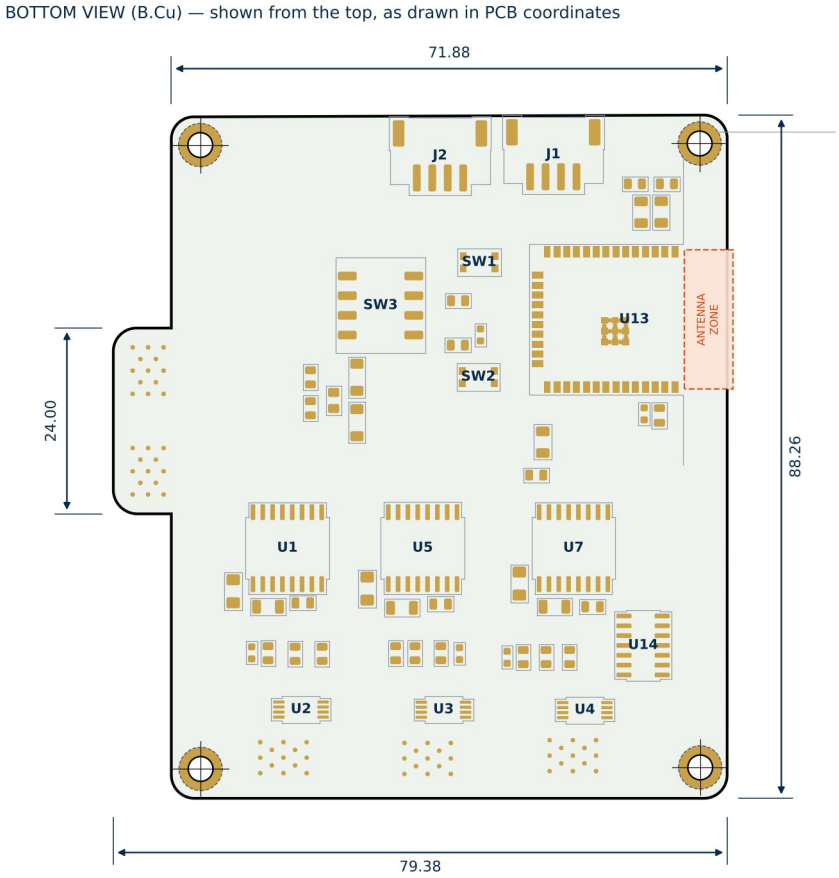


Figure 4b — Bottom view dimensions and component outlines (as drawn in PCB coordinates)

7.2 Mounting holes

Ref	X (mm)	Y (mm)	Hole	Pad	Net
H1	11.24	84.39	Ø3.2 mm (M3, DIN 965 head)	Ø5.6 mm	Not connected (isolated)
H2	75.82	84.59	Ø3.2 mm (M3, DIN 965 head)	Ø5.6 mm	Not connected (isolated)
H3	75.89	3.99	Ø3.2 mm (M3, DIN 965 head)	Ø5.6 mm	Not connected (isolated)
H4	11.29	3.79	Ø3.2 mm (M3, DIN 965 head)	Ø5.6 mm	Not connected (isolated)

Nominal hole pattern 64.6 × 80.6 mm (hole centres are within ≈ 0.2 mm of a perfect rectangle in the source layout). Holes are plated with an isolated annular pad, so metal standoffs do not short to GND.

7.3 Elevations

Silhouettes use nominal package heights (see §8); verify against your selected parts, especially capacitors, inductors and the JST connectors.

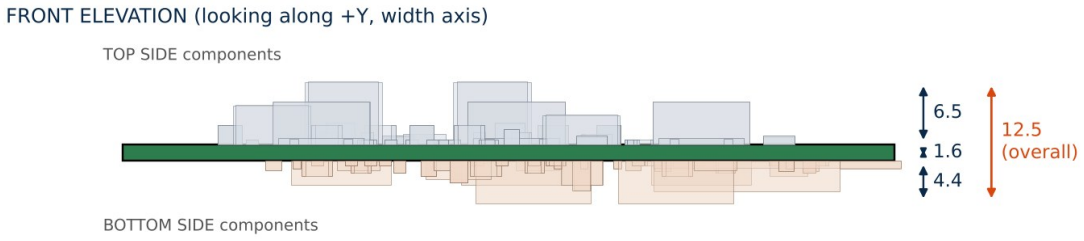


Figure 5a — Front elevation

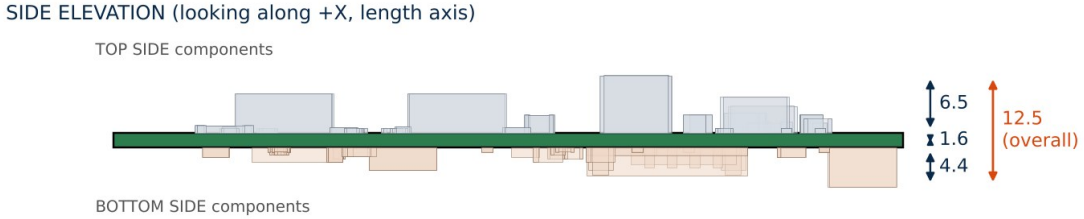


Figure 5b — Side elevation

8 3D views

Renders are generated directly from the PCB layout: board texture (copper, mask, pads, silkscreen) with extruded component bodies at nominal package heights. They are illustrative and do not use manufacturer STEP models.

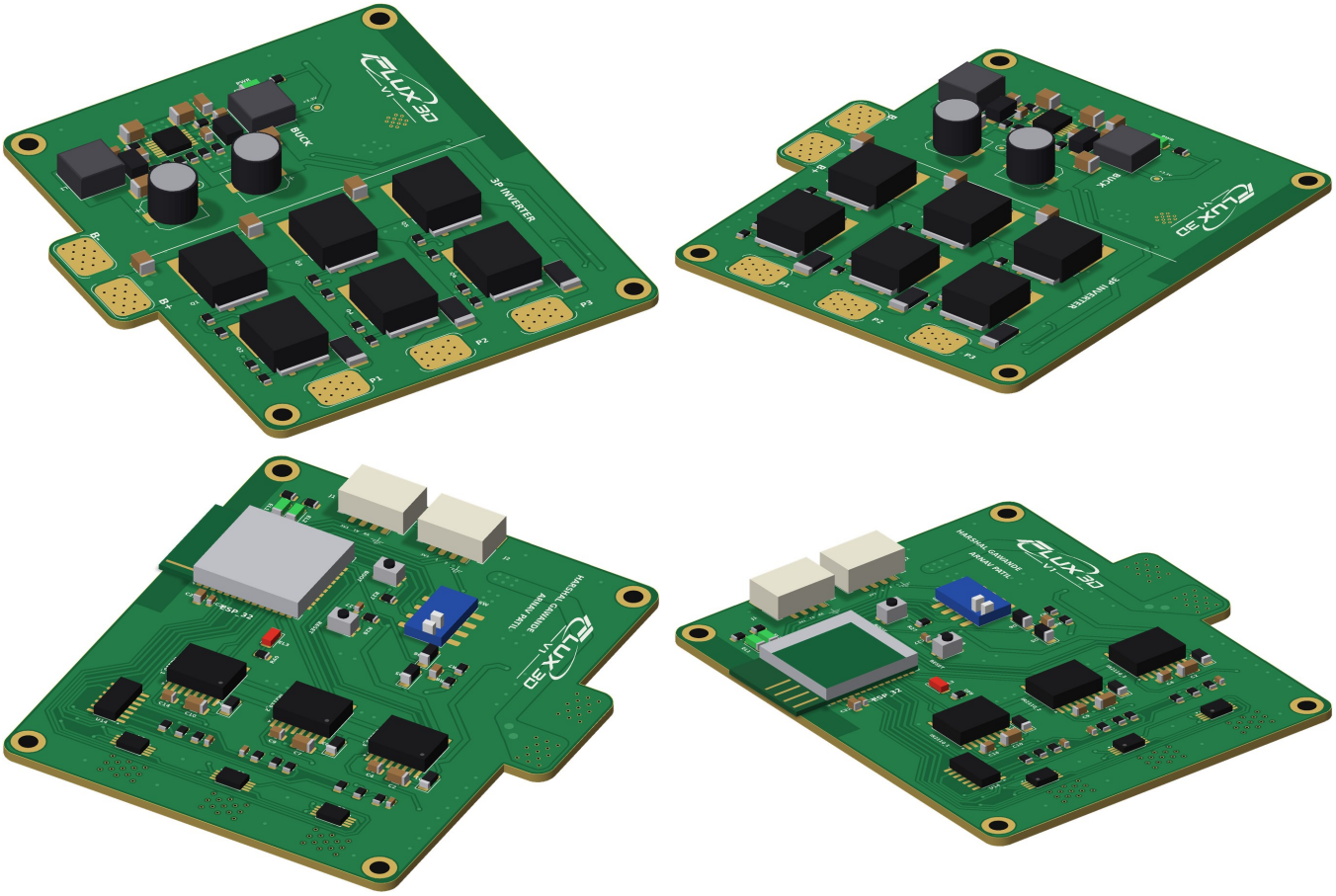


Figure 6 — Top side (upper row) and bottom side (lower row), two viewing angles each

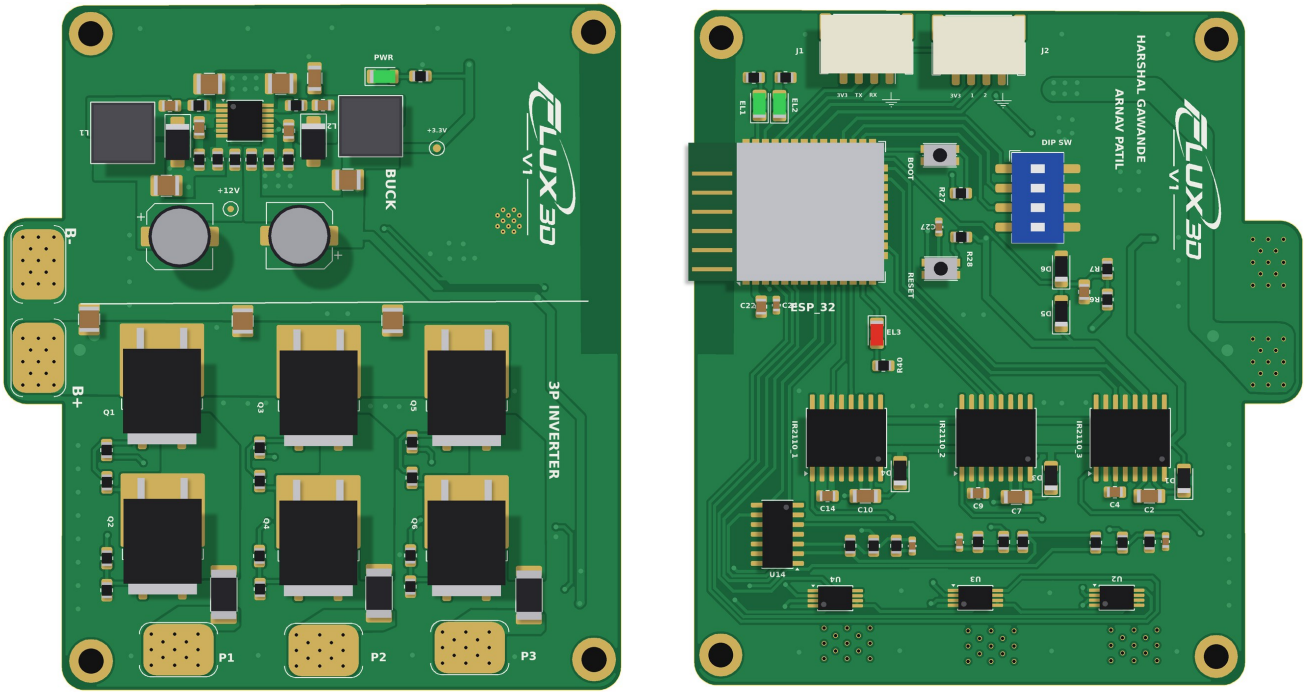
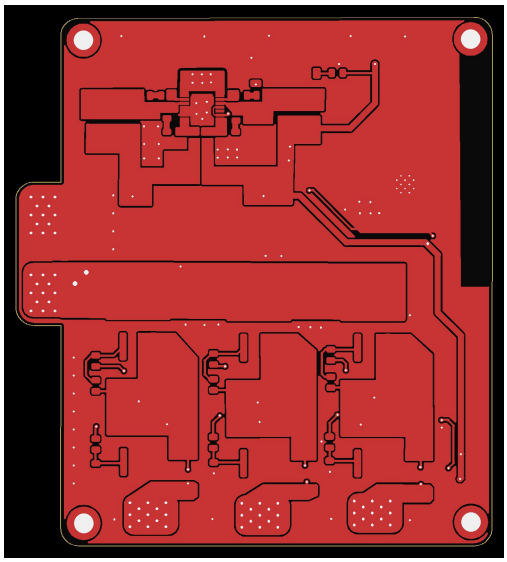
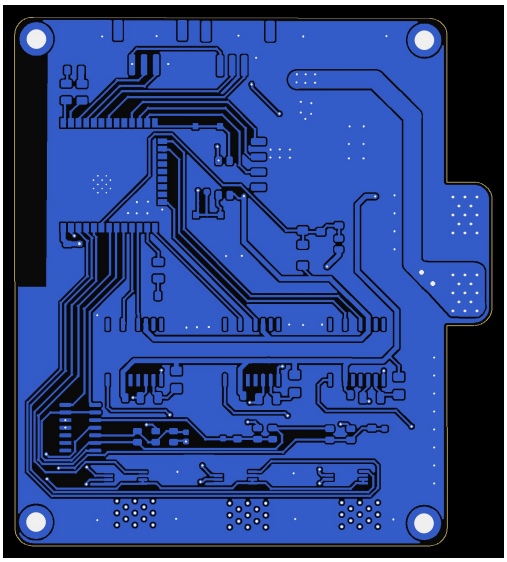


Figure 7 — Plan views: top side (left), bottom side (right, mirrored)

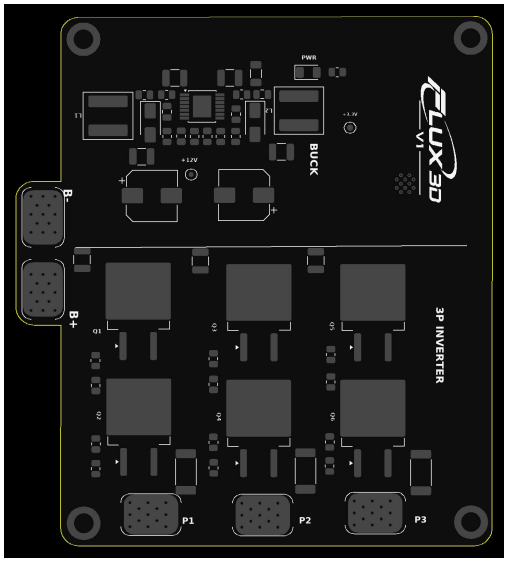
9 PCB layout and fabrication data



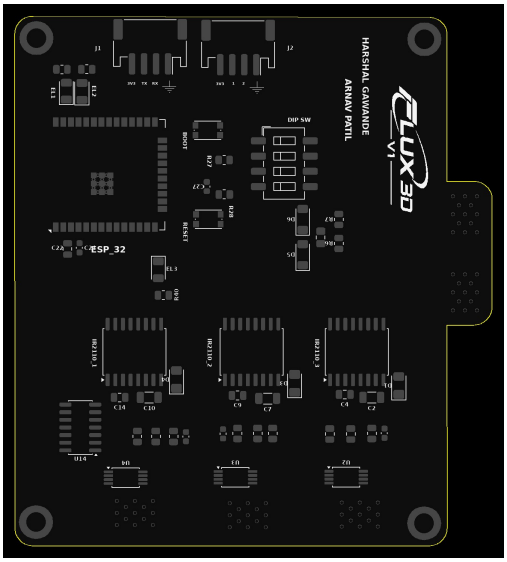
Top copper (F.Cu)



Bottom copper (B.Cu, viewed from below)



Top silkscreen + pads



Bottom silkscreen + pads (viewed from below)

9.1 Fabrication summary

Item	Specification	Source
Layer count / thickness	2 copper layers (F.Cu, B.Cu) · 1.6 mm	PCB file
Copper weight / finish / mask colour	Not specified in the design files — confirm with the fabricator (2 oz outer copper and ENIG/HASL recommended for the power path)	—
Solder mask	Vias tented on both sides	PCB setup
Minimum track / clearance	0.20 mm / 0.20 mm (default net class); copper-to-edge 0.50 mm; hole clearance 0.25 mm	Project rules
Track widths used	0.2 · 0.4 · 0.8 · 1.0 mm; 3.0 mm bottom power tracks	PCB
Vias	0.6 / 0.3 mm × 97 · 0.8 / 0.4 mm × 1 · 1.5 / 0.75 mm × 2 (100 total)	PCB
Drill sizes	0.2 mm × 12 (ESP32 thermal-pad vias) · 0.3 mm × 97 · 0.4 mm × 66 · 0.75 mm × 2 · 3.2 mm × 4 (plated)	PCB
Placement	61 components top, 50 bottom (111 footprints incl. pads, test pads and holes)	PCB
Routing	717 track segments (207 top, 510 bottom); copper zones on both layers; ESP32 antenna keep-out on top layer	PCB
EDA	KiCad pcbnew 10.0 (file version 20260206)	PCB file

Note: the smallest drill (0.2 mm, 12 holes in the ESP32 module ground-pad footprint) is below the project's own 0.3 mm minimum through-hole rule; check that your fabricator supports it or enlarge those holes.

10 Bill of materials

Ordering BOM from the project (ordering supplier details, prices and spares omitted here). “Qty” is the number of placements on one board.

#	Designators	Qty	Value / function	Package	Part (manufacturer)
1	Q1–Q6	6	NMOS 75 V 80 A	TO-263-2 (D2PAK)	IRFS3607TRLPBF (Infineon)
2	U1, U5, U7	3	IR2110S	SOIC-16W	IR2110STRPBF (Infineon)
3	D1, D3, D4	3	Bootstrap diode (sch: "uf4007")	D_1206 HandSolder pads	ES1J SOD-123FL (HXY)
4	C2, C7, C10	3	1 µF (bootstrap)	1206	CL31B105KBHNFNE (Samsung)
5	C4, C9, C14	3	1 µF (sch: "C", no value)	0805	CC0805JKX7R9BB105 (Yageo)
6	R14, R15, R19, R20, R23, R24	6	22 Ω (gate)	0805	0805W8J0220T5E (Uniohm)
7	R16, R17, R21, R22, R25, R26 (+ others)	11	10 kΩ	0805	RC0805FR-0710KL (Yageo)
8	C12, C13	2	10 µF (VBUS)	1210	C3225X7S1H106KT000N (TDK)
9	C23, C25, C26	3	2.2 µF (commutation)	1210	CS3225X7R225K101NRK (Samwha)
10	R1, R2, R3	3	Shunt (sch: "R_US", no value)	2512	HOYLR2512E-3W-1mR
11	U2, U3, U4	3	INA240A1PW	TSSOP-8	INA240A1PWR (TI)
12	U14	1	LMV324	SOIC-14	LMV324IDT (ST)
13	R29, R30, R31, R33, R35, R37	6	33 kΩ	0805	RC0805FR-0733KL (Yageo)
14	C28, C29, C30	3	1 nF	0603	CL10C102JB8NFNC (Samsung)
16	R6	1	56 kΩ	0805	WR08X5602FTL (Walsin)
17	R7	1	4.7 kΩ	0805	RC0805FR-074K7L (Yageo)
18	C11	1	100 nF	0805	CL21B104KBCNNNC (Samsung)
19	D5, D6	2	Clamp Schottky (sch: TMMBAT45)	MiniMELF	TMMBAT42FILM (ST)
20	U6	1	TPS54383PWPR	HTSSOP-14 (PowerPAD)	TPS54383PWPR (TI)
21	L1	1	47 µH (12 V rail)	XAL7050 footprint (7.5×7.2)	SWPA8040S470MT (Sunlord)
22	L2	1	22 µH (3.3 V rail)	XAL7050 footprint	ASPI-0630LR-220M-T15 (Abracon)
23	D2, D7	2	SS34	SMA	SS34 (Slkor)
24	R10	1	51.1 kΩ	0805	AC0805FR-0751K1L (Yageo)
25	R11	1	3.65 kΩ	0805	RC0805FR-073K65L (Yageo)
26	R8	1	20 kΩ	0805	RC0805FR-0720KL (Yageo)
27	R9	1	6.34 kΩ (sch: "6.4k")	0805	AC0805FR-076K34L (Yageo)
28	R5, R12	2	3.3 Ω	0805	RC0805FR-073R3L (Yageo)
29	R4, R13	2	10 Ω	0805	0805W8F100JT5E (Uniohm)
30	C5, C17	2	1 nF	0805	CL21C102JBCNNNC (Samsung)
31	C6, C16	2	33 nF	0805	TCC0805X7R333K500DT (CCTC)
32	C8	1	4.7 µF	1206	CL31B475KAHNFNE (Samsung)
33	C1, C20	2	10 µF	1210	CS3225X7R106K250NRK (Samwha)
34	C3	1	47 µF (12 V bulk)	CP_Elec_8x10 (Ø8 mm)	VZH470M1VTR-0806 (Lelon)
35	C19	1	100 µF (3.3 V bulk)	CP_Elec_8x10 (Ø8 mm)	VEJ101M1ETR-0806 (Lelon)
36	U13	1	ESP32-WROOM-32E	Module	ESP32-WROOM-32E-N4 (Espressif)
37	C22	1	10 µF	0805	TMK212BBJ106KGHT (Taiyo Yuden)
38	C24	1	100 nF	0603	TCC0603X7R104K500CT (CCTC)
39	C27	1	1 µF	0603	GRM188R71E105KA12D (Murata)
40	SW1, SW2	2	Tactile (BOOT / EN)	C&K; KMR2	KMR221NGLFS (C&K;)
41	SW3	1	4-pos DIP	SMD 2.54 mm, 6.7×11.72, low-profile	SDA04H1SBDR (C&K;)
42	D9, D10, D8	3	Green LED	1206	LTST-C150KGKT (Lite-On)
43	D11	1	Red LED	1206	LTST-C150KRKT (Lite-On)
44	R18, R38, R39, R40	4	330 Ω (sch: R38–40 "R_US")	0805	RC0805FR-07330RL (Yageo)
45	J1, J2	2	JST-PH 4-pin, SMD horizontal	S4B-PH-SM4-TB	WAFER-PH2.0-4PWB (XUNPU)

Not on the BOM (no parts to buy): TP1, TP2 test pads; H1–H4 M3 mounting holes; U8–U12 solder wire pads (VBUS, GND, phases A/B/C). Not included: PCB, battery and motor wire, battery connector, bulk electrolytic capacitor for VBUS.

11 Design review notes and known issues

Items found while analysing the schematic, netlist and BOM. Items 1 and 7 are from the netlist analysis performed for this datasheet; items 2–6 and 8 are from the project's BOM design-review sheet. None of these has been verified on hardware — please check them against your schematic before ordering PCBs or powering a build.

#	Severity	Issue	Detail / suggested action
1	**HIGH**	Debug LEDs D9, D10, D11 appear reverse-biased	In the netlist each LED cathode goes to an ESP32 GPIO (LED3, LED2, LED1) and each anode goes through a 330 Ω resistor (R38, R39, R40) to **GND**, so the LED can never conduct. Fix: mount the LED rotated 180° (anode to GPIO, cathode via resistor to GND), or correct the schematic (LED polarity) in the next revision. D8 (PWR) is wired correctly.
2	**HIGH**	BEMF dividers exceed LMV324 input range	R33/R35/R37 (33 kΩ) over R32/R34/R36 (10 kΩ) give 25.2 × 10/43 = 5.9 V at full 6S, but U14 runs from 3.3 V, so the comparators saturate or clamp above roughly 14 V bus and the inputs are over-stressed. Change the bottom resistors to 3.3 kΩ (→ 2.3 V at 25.2 V) or otherwise re-scale the network.
3	**HIGH**	Bootstrap diode value vs footprint	D1/D3/D4 are labelled "uf4007" on a 1206 hand-solder footprint. The BOM substitutes ES1J (SOD-123FL, 600 V, 35 ns); consider changing the footprint to match.
4	MED	No local bypass on INA240 (U2–U4) or LMV324 (U14)	No capacitor is connected to their 3.3 V pins in the netlist. Add 100 nF at each supply pin.
5	MED	TPS54383 input headroom	VIN abs. max is 30 V (28 V recommended). A fully charged 6S pack is 25.2 V; regen or switching spikes may exceed this. Add a TVS (e.g. 30 V class) and/or an input filter near U6.
6	LOW	Missing / mistyped values in schematic	C4, C9, C14 ("C"), R1–R3, R38–R40 ("R_US") have no value; C10 is "iuf"; R9 is "6.4k" (not an E96 value; 6.34 kΩ gives 3.32 V). Write the BOM values back to the schematic.
7	LOW	ADC2 and strapping-pin usage	Bemf2 (GPIO25) is on ADC2, which cannot be sampled while Wi-Fi is active (use it as a digital comparator input, or move it to an ADC1 pin). GPIO0/2/12/15 are strapping pins shared with BOOT and gate-driver inputs.
8	LOW	Shunt sizing / inductor package	1 mΩ gives ≈ 40 mA/LSB; a 2 mΩ shunt doubles resolution if peak current is below 40 A. The BOM lists an 8 × 8 mm SWPA8040 for L1 (47 µH) on the XAL7050 footprint (≈ 7.7 × 8.0 mm body outline) — check fit; 33 µH SWPA6045S330MT is the in-stock alternative noted in the BOM.

12 Revision history and notices

Revision	Date	Description
Rev 1.0 (Preliminary)	8 November 2026	First release, generated from bldc_controller_v1 (.kicad_pcb / .kicad_sch / .kicad_pro) and BLDC_Controller_v1_BOM_Robu.xlsx.

Creators / owners. Harshal Gawande and Arnav Patil.

Source data. Dimensions, coordinates, pin/net assignments, drill and track data are read from the KiCad PCB file; schematic values and part numbers from the KiCad schematic and BOM. **Creators / owners:** Harshal Gawande and Arnav Patil (names also printed on the PCB silkscreen, with a "V1" logo).

Disclaimer. This preliminary datasheet is provided for information only. Electrical limits are derived from component ratings and calculations, not from testing. No warranty is given; the user is responsible for verifying the design, thermal performance, safety (including battery protection) and regulatory compliance before use. High-current battery systems can cause fire and injury if mis-wired or unprotected.